

NI Circuit Design Suite For All Your Circuit Designing Needs

The NI Circuit Design Suite combines NI Multisim and Ultiboard software for circuit design, simulation, validation and layout platform. In this month's DVD with EFY Plus, we present to you NI Circuit Design Suite 13.0, which can be used for 45 days for all your circuit designing needs

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National Instruments Circuit Design Suite is a suite of electronics design automation (EDA) tools having Multisim and Ultiboard, which can be installed from the accompanying DVD. Multisim is the schematic capture and simulation program designed for schematic entry, simulation and linkage to steps such as PCB layout. It also includes mixed analogue/digital simulation capability and microcontroller co-simulation. Ultiboard is for designing printed circuit boards, doing some basic mechanical CAD operations and preparing the PCBs for manufacture.

Features of Multisim

Let us take a look at the main features of Multisim.

Learn analogue circuits with circuit parameters. Circuit parameters provide the ability to define schematic-level variables that can then be easily modified to match homework, lab problems and other parametric analysis of a circuit. They allow us to highlight the key components and variables that characterise the response of analogue and power circuits. Prior to Multisim 13.0, the calculation of important variables and setting of related component values had to be done manually.

Variables embedded within a schematic can be based on theoretical calculation, which cascade to all applicable components. These variables

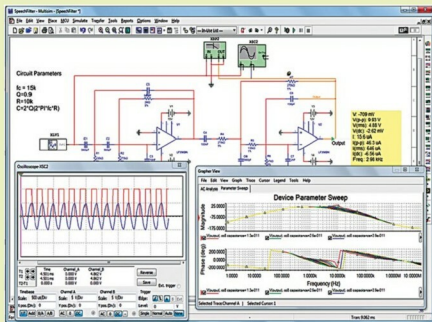


Fig. 1: Multisim simulation environment (Courtesy: www.ni.com)

Benefits of using NI Circuit Design Suite

You can:

- Capture and simulate circuit designs with NI Multisim software
- Easily design circuits with intuitive and cost-effective design tools
- Quickly improve circuit behaviour with interactive SPICE simulation
- Seamlessly transition to NI Ultiboard for PCB layout and routing
- Reduce design errors with leading analogue simulation capabilities

can also be used in expressions such that specific circuit characteristics can be determined dynamically, based on other component values within the schematic. This allows students and educators to bridge the gap between theory and practical, as well as opens up the possibility of far more advanced simulation analysis.

Better design performance with circuit parameters. Circuit parameters add sophistication to any simulation approach by allowing an engineer to easily explore different scenarios for various circuit structures and component values. The parameters also help in performing advanced analyses such as temperature sweep, parametric